problem statement

Proposition 0.1. *Define $g_{n,i}(x) = I_x(i, n-i+1)$, we want to prove $g^{-1}_{n,n-j}(x)^n$ is an increasing function with respect to n, or $g_{n,n-j}(x^{\frac{1}{n}})$ is a decreasing function with respect to n. We have known $g_{n,n-j}(x)$ is a decreasing function with respect to n and $g^{-1}_{n,n-j}(x)$ is an increasing function with respect to n.*